

12 (**long**) Consider the mass–spring–dashpot system shown in Fig. 8, where  $f_1$  is an applied force,  $m_1$  and  $m_2$  are the masses of the bodies,  $k_1$  and  $k_2$  are spring stiffnesses, and  $\lambda$  is the dashpot viscosity.

(a) The equation of motion can be expressed in the form

$$\mathbf{M}\ddot{\mathbf{y}} + \mathbf{C}\dot{\mathbf{y}} + \mathbf{K}\mathbf{y} = \mathbf{f},$$

where the vector  $\mathbf{y}$  contains the displacement of each body. Formulate the matrices  $\mathbf{M}$ ,  $\mathbf{C}$  and  $\mathbf{K}$ , and the vector  $\mathbf{f}$ . [6]

(b) For the case  $\lambda = 0$ , find the natural frequencies of the system when  $k_1 = k_2$ . [8]

(c) For the case  $\lambda = 0$ , determine the mode shapes when  $k_1 = k_2$ . [6]

(d) Consider the case  $\lambda = 0$ ,  $k_1 = k$ ,  $k_2 = 2k$ ,  $m_1 = m_2$  and  $f_1 = 0$ . The springs are unstretched when  $\mathbf{y} = [0, 0]^T$ . For the initial conditions  $\mathbf{y}_0 = [1, 1]^T$  and  $\dot{\mathbf{y}}_0 = [0, 1]^T$ , find an expression for the response of the system as a function of time. [10]

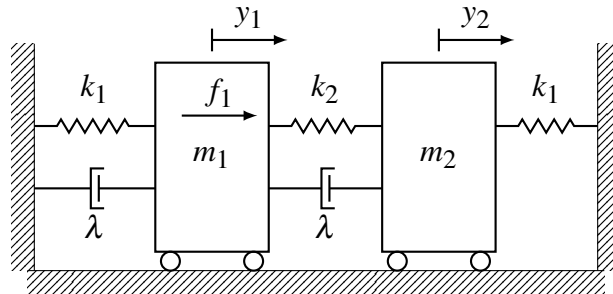


Fig. 8

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